Machine Learning-Based Buyer Segmentation and Investment Profiling for Real Estate Market Intelligence

Parcl Co. Limited | Unified Mentor

Research Paper and Technical Implementation Report

Prepared using the uploaded customer and property datasets

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Abstract

This project uses the uploaded client and property datasets to build an end-to-end buyer segmentation workflow. It joins 2,000 unique clients to 7,305 sold property records, engineers transaction features, applies K-Means clustering and compares the result with Ward hierarchical clustering. The selected K is 3 with a silhouette score of 0.148. This modest score means clusters overlap and should be treated as exploratory profiles rather than definitive personas.

1. Introduction and Problem Statement

Real estate companies serve buyers with different purchase purposes, financing needs, geographic locations and transaction histories. The project aims to identify patterns in buyer characteristics and property purchase activity to support more informed market intelligence, customer communication and marketing analysis.

2. Objectives

- Clean and validate customer and property datasets.
- Join client records with completed sold transactions.
- Derive age and buyer-level purchase indicators.
- Encode categorical features and standardize numeric features.
- Train K-Means and evaluate candidate cluster counts with inertia and silhouette score.
- Use Ward hierarchical clustering as a structural comparison.
- Build an interactive Streamlit dashboard with the four requested modules and filters.

3. Dataset Description

The customer file contains 2,000 rows and fields for client ID, client type, date of birth, gender, country, region, acquisition purpose, satisfaction, loan application and referral channel. It also contains first and last names, which are excluded from modeling and analytic exports. The property file contains 10,000 listing records with transaction dates, unit categories, floor areas, sale prices, listing status and client references.

The datasets are joined using clients.client_id = properties.client_ref. Only Sold listings with a client reference are counted as completed purchases. There are 7,305 such records, and all 2,000 clients have at least one linked sold property in this extract.

4. Data Cleaning and Feature Engineering

- Remove duplicate client IDs and duplicate listing IDs.
- Trim categorical labels and handle missing values.
- Parse date of birth and derive age using 31 October 2026 as the analysis reference date.
- Parse currency-formatted sale price to numeric values.
- Aggregate sold property count, total purchase value, average purchase price, total/average floor area, apartment and office counts, and office share per buyer.
- Exclude names and identifiers from the feature matrix; client_id is retained only as a key.

5. Feature Encoding and Scaling

Categorical features are client_type, country, region, acquisition_purpose, loan_applied and referral_channel. They are one-hot encoded. Numeric features are age, satisfaction score, properties purchased, log total purchase value, log average purchase price, total floor area and office share. Numeric missing values are median-imputed and standardized with StandardScaler. Gender is not used as a clustering feature.

6. Clustering and Evaluation

K-Means is trained for K values from 2 through 8 with random_state=42 and n_init=20. Inertia is recorded for the elbow curve and silhouette score is used as the primary selection guide. The selected result is K=3. The silhouette score is 0.148, which indicates modest separation; the cluster solution should not be treated as a definitive segmentation.

Ward hierarchical clustering is applied to a deterministic sample of up to 500 buyers. The adjusted Rand index between the K-Means and hierarchical assignments on that sample is 0.353. This is a validation aid, not the model used for final cluster assignment.

7. Exploratory Data Analysis

- Unique clients after deduplication: 2,000.
- Property records: 10,000; sold records: 7,305; available listings: 2,695.

- Investment-purpose clients: 615 (30.8%); home-purpose clients: 1,385 (69.2%).
- Loan applicants: 736 (36.8%).
- Mean satisfaction: 3.03/5; mean age: 56.4 years.
- Total value of linked sold transactions: $2,520,750,961; average sold listing price: $345,072.
- Buyers represented across 10 countries and 57 regions.

8. Cluster Profiles

Cluster	Buyers	Avg age	Investment	Loan	Avg properties	Avg total value	Avg unit price
0	55	65.6	32.7%	36.4%	7.11	$2,383,987	$339,913
1	1035	55.8	29.0%	35.9%	3.72	$1,440,427	$392,627
2	910	56.6	32.6%	37.8%	3.37	$987,681	$295,731

Cluster 0 is a small higher-activity purchaser group with the highest average property count and total linked purchase value. Cluster 1 has the highest average unit purchase price. Cluster 2 has the lowest average unit purchase price and average total linked purchase value. Investment and loan shares are relatively similar across clusters.

9. Streamlit Application

- Buyer segmentation overview: cluster distribution, elbow curve and silhouette scores.
- Investor behaviour: acquisition purpose, loan application, referral channel and purchase value by cluster.
- Geographic analysis: country choropleth, region/cluster treemap and country-region counts.
- Segment insights: descriptive profile table and buyer-level records.
- Filters: country, region, acquisition purpose and client type; filtered CSV download.

10. Business Recommendations

- Review higher-activity buyers for relationship management and portfolio servicing, after business validation.
- Use purchase-price and purchase-volume patterns to inform property recommendation journeys.
- Compare referral-channel mix across clusters to assess acquisition source patterns.
- Review geographic distributions before localized campaign planning.
- Use loan application only as a financing-interest indicator, not as a creditworthiness measure.
- Validate proposed actions with business teams and monitor segment stability over time.

11. Limitations and Responsible Use

- The silhouette score is modest; clusters overlap and are exploratory.
- The supplied data does not contain verified income, net worth, investor accreditation, first-time-buyer status or luxury status; those personas cannot be confirmed.
- The client_type category is Company in the data, not Corporate.
- Gender is excluded from model features. Age is included for demographic profiling and must be reviewed for lawful and fair use.
- Clustering is not causal and must not be used for discriminatory decisions about housing access, pricing, credit or eligibility.
- Geographic visualization is country-level because region-level latitude and longitude are not supplied.

12. Conclusion

The project implements a complete buyer segmentation workflow using the supplied client and property datasets. It combines customer attributes with completed property transaction aggregates, applies K-Means and hierarchical comparison, evaluates cluster counts and presents the results in Streamlit. The three-cluster solution reveals differences in transaction activity and purchase values, but its modest silhouette score means that the result should be used as an exploratory market-intelligence aid and validated with business stakeholders.

References

- MacQueen, J. (1967). Some Methods for Classification and Analysis of Multivariate Observations. Proceedings of the Fifth Berkeley Symposium on Mathematical Statistics and Probability.

- Rousseeuw, P. J. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics, 20, 53–65.
- Ward, J. H. (1963). Hierarchical Grouping to Optimize an Objective Function. Journal of the American Statistical Association, 58(301), 236–244.
- Pedregosa, F. et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.